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LC 1011: Capacity to Ship Packages Within D Days

Goldman Sachs

Flipkart

Agoda

Problem Statement & Requirements



there is a deadline to ship packages within d days, each day we can transfer weights, but total weight must not exceed ship capacity

return minimum possible capacity

Intuition & Logic

lets say you found out that a ship with capacity 100 was able to do the task, then 101, 102, ---- will also do

hence seeing the (monotonic + minimum) → **binary search**

Binary Search Bounds

$l = \text{max_element}()$

must carry at least heaviest package

$h = \text{accumulate}()$

ship everything in one day

C++ Implementation

```
class Solution {
public:

    bool check(vector<int>& weights, int days, int cap){
        int currday = 1;
        int currw = 0;

        for(int w : weights){
            if(currw + w <= cap){    //agar ye weight daala ja
                                   skta he to daal do
                currw += w;
            }

            else{    //agar nhi daal skte to next day issi weight
                    ke saath start krenge
                currday++;
                currw = w;
            }
        }

        if(currday <= days){
            return true;
        }
        else{
            return false;
        }
    }

    int shipWithinDays(vector<int>& weights, int days) {

        int n = weights.size();
        int l = *max_element(weights.begin(), weights.end());
        int h = accumulate(weights.begin(), weights.end(), 0);

        int ans = h;
        while(l <= h){
            int mid = l + (h - l) / 2;

            if(check(weights, days, mid)){
                ans = mid;
            }
        }
    }
};
```

```
        h = mid - 1;
    }

    else{
        l = mid + 1;
    }
}

return ans;
}

};
```